



Workshop #1

Mastercard Data Challenge 2026

Turning Data into Insight
(Inclusive Growth Score / IGS)



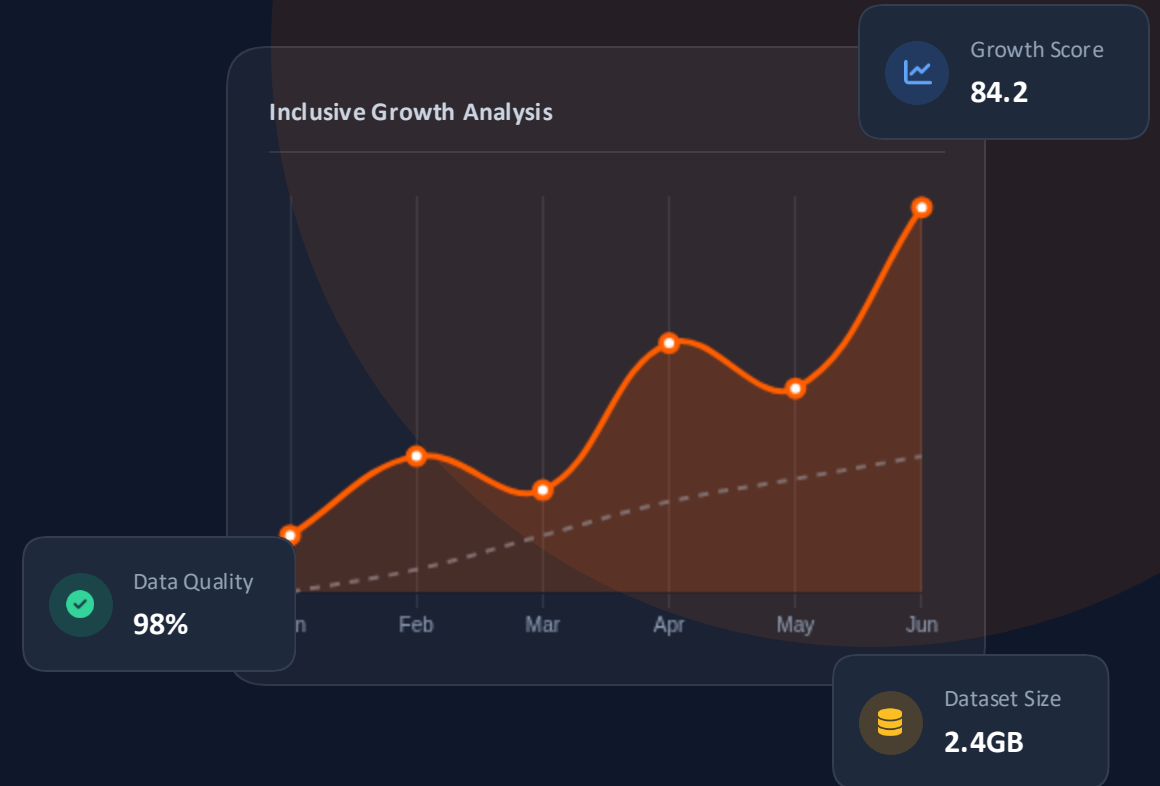
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February 19, 2026

Atlanta University Center



Instructors

 Spelman College



Dr. Margie Ruffin

Assistant Professor of Computer Science

Areas of Expertise



Data Collection



Data Cleaning



CS Education

 Morehouse College
College



Dr. Dewayne Dixon

Visiting Assistant Professor of Computer Science

Areas of Expertise



Statistical Modeling



Interpretable ML



AI Ethics

Session Overview

90 Minute Workshop



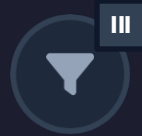
Setting the Stage

Context & Goals



Data Sourcing

IGS & Public
Data



Data Analysis

Cleaning & EDA



Data Visualization

Patterns & Stories



Statistical Modeling

Drivers & Insight



Q&A

Next
Steps



Workshop Format

Each section will be followed by a brief Q&A and a debrief as a collective.

2026 Data Challenge: Setting the Stage

🎯 Challenge: Find, evaluate, and propose a small business solution for communities with an **IGS < 45**.



Place

Environment & Access

Housing availability & cost
Transit accessibility
Essential services
Infrastructure quality
Digital access (broadband)

"Physical conditions that shape opportunity."



Economy

Local Economic Activity

Job availability & growth
Consumer spending
Investment flow
Income levels
Small business activity

"Strength and growth of business activity."



Community

Social Inclusion & Well-Being

Health access
Education quality
Social capital
Financial inclusion
Community stability

"Residents' ability to participate and benefit."



Reminder: IGS is a community-level signal. Treat it as context, not a label for individuals.
individuals.

Sourcing the Right Data



Mastercard Inclusive Growth Score (IGS)

Primary Tool

- 1 Step-by-step walkthrough (Posted in Canvas)
- 2 Analyze Census Tract Views
- 3 Apply Filters & Benchmarks
- 4 Export Results



Supporting Public Datasets

Contextual Data



Census / ACS

Income, education, housing, demographics



BLS

Employment stats, labor participation rates



FCC

Broadband access and digital connectivity maps



CMS / Health

Local public health access proxies



IRS / Registries

Business activity and small business proxies



ICPSR / Reports

Qualitative sources and local housing/policing data



Workshop Note

We will demo techniques using a synthetic IGS dataset (fictional communities) so everyone starts with the same clean baseline

Sourcing the Right Data: Opportunity Zones



What is an Opportunity Zone?

A **federally designated low-income or distressed geographic area** where investments may qualify for tax incentives intended to promote economic development and job creation.



Use as one lens for prioritizing or comparing areas (not a substitute for the full IGS story).

For more detailed information:

[IRS FAQ: Opportunity Zones Frequently Asked Questions](#)

Data Cleaning and Analysis

> Python Workflow Demo (Pandas)

● ● ● analysis_workflow.py

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Load exported IGS data
df = pd.read_csv("mastercard_igs_synthetic_data.csv",
                 parse_dates=["Date"])

# 2. Inspect structure
df.info()
df.head()

# 3. Check data quality
print(df.isna().sum()) # Missing values
print(df.describe()) # Summary stats

# 4. Validate keys
unique_tracts = df['tract_id'].nunique()
```



Common Data Quality Issues

- Missing values across key indicators
- Inconsistent geographic identifiers (e.g., Tract IDs)
- Different time periods across variables
- Outliers or extreme values affecting averages
- Duplicate records and formatting inconsistencies



Structuring for Analysis

- Standardize column names & units for consistency
- Align census tract identifiers across all sources
- Handle missingness strategies: Drop, Impute, or Flag
- Generate summary statistics and distributions

Example Processes

ETL Pipeline



Source Data

IGS CSV + Census API



Transform

Clean, Filter,
Normalize



Clean Dataframe

Ready for
Analysis



Visual Output

Plots &
Insights

Data Merge Strategy



IGS Data

Key: Tract_ID



ACS Data

Key: GEOID



Inner Join
Join



Unified Dataset

Combined metrics for
modeling

Geospatial Mapping



Shapefile

Tract Boundaries
Boundaries

+ Geometry



Metric Data

Scores & Values



Choropleth Map

Spatial Visualization
Visualization

From Analysis to Visual Storytelling



Analysis → Potential Stories

Finding the narrative in data

- ❓ Which tracts/areas have the the lowest inclusive growth? growth?
- 📊 Which dimension drives low scores (Place/Economy/Community)?
- ≠ Where is growth happening happening without inclusion?
- 📈 inclusion? Which areas improve or decline over time?



Solidifying a Data Story

Structuring the insight

- 🎯 Identify a meaningful pattern pattern or disparity
- ✓ Validate with benchmarks and comparisons
- 🌐 Connect pattern to real-world context
- 💬 Communicate implications (policy, investment, equity)



Tools & Workflows

Execution stack

Python / R

Cleaning,
EDA,
modeling

Excel

Quick
inspection +
validation

Tableau/ PowerBI

Dashboards + spatial views

Documentation

Reproducible workflow

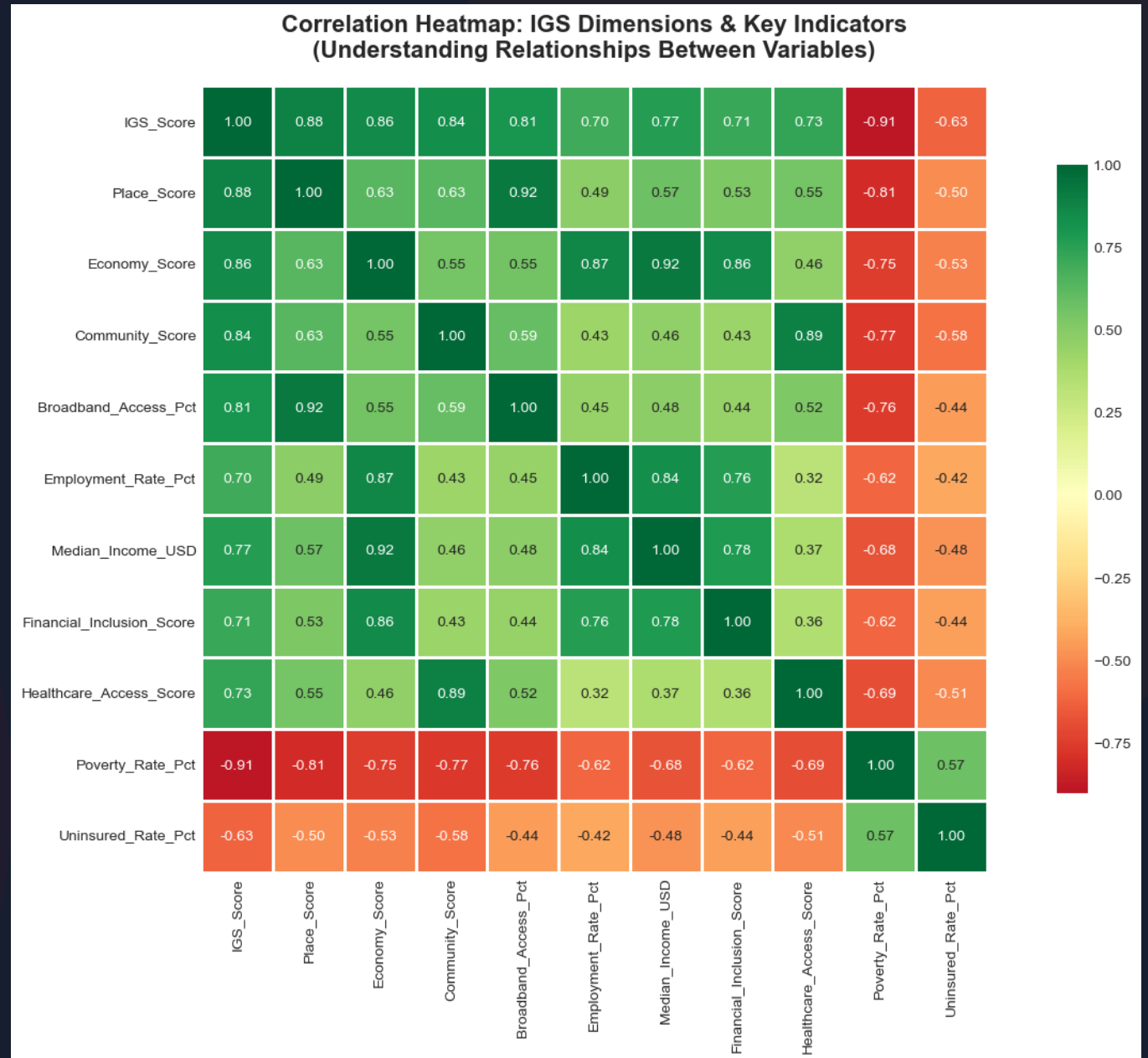
Correlation Heatmap

What it shows: How strongly different variables are related to each other.

How to read it:

- Dark green = strong positive correlation (when one goes up, the other goes up)
- Dark red = strong negative correlation (when one goes up, the other goes down)
- Light colors = weak or no correlation

Why it matters: Helps us understand which factors move together and might influence IGS.



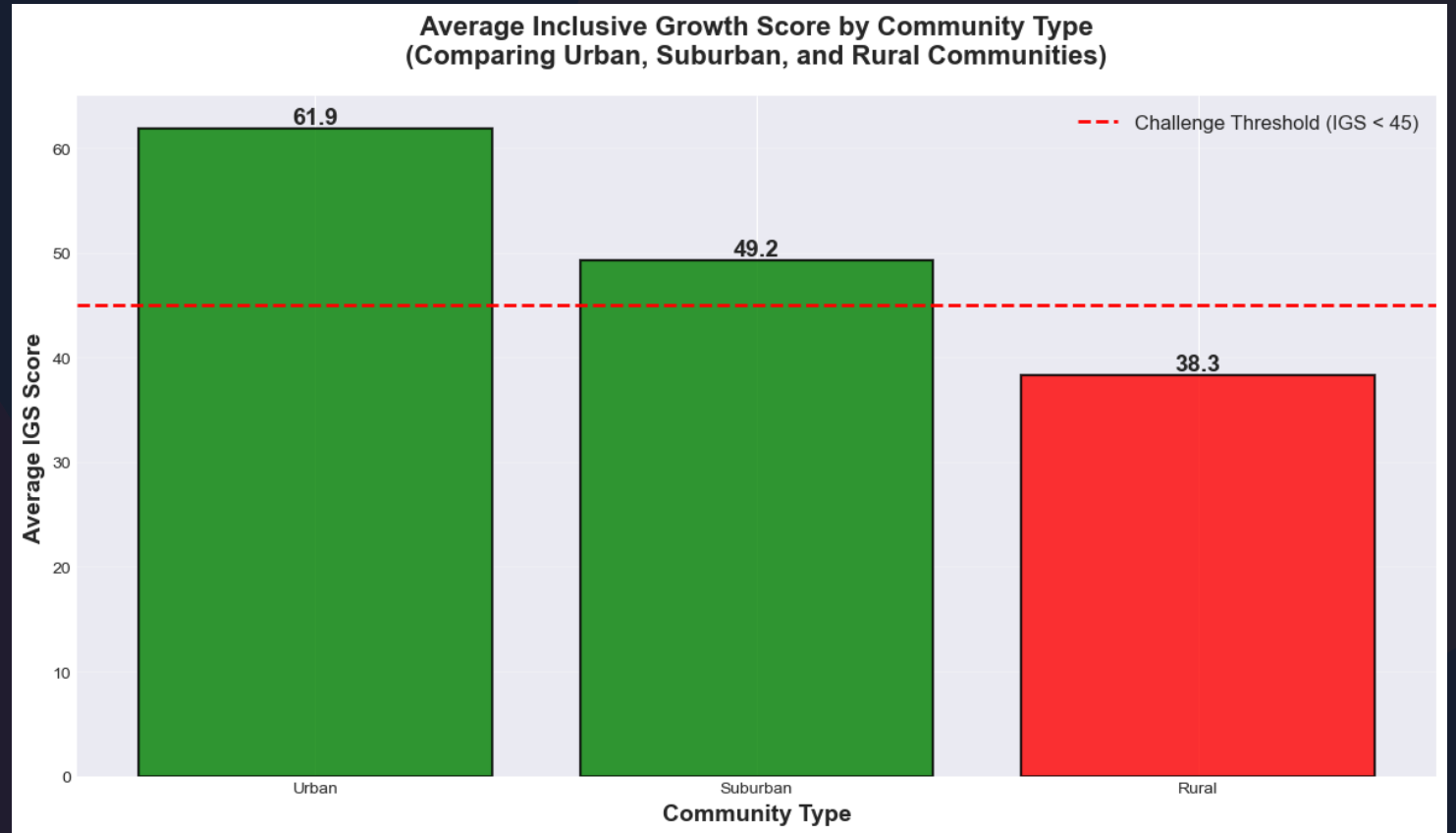
Bar Chart

What it shows: Average IGS scores for Urban, Suburban, and Rural communities.

How to read it:

- Height of bars = average IGS score
- Red dashed line = challenge threshold (45)
- Color = whether community type is above/below threshold

Why it matters: Shows which types of communities need more support.



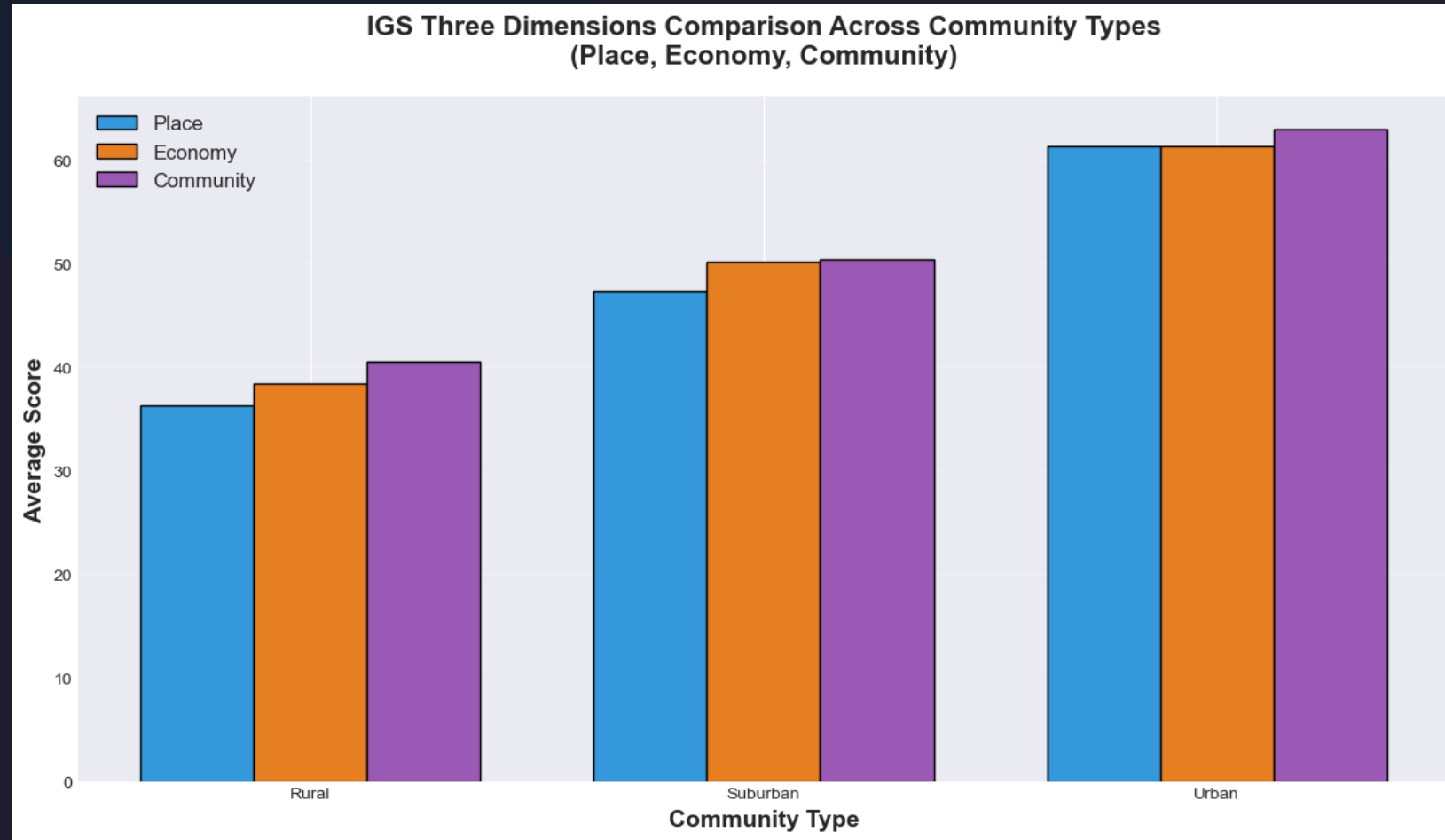
Grouped Bar Chart

What it shows: How Place, Economy, and Community scores differ across community types.

How to read it:

- Each community type has 3 bars (Place, Economy, Community)
- Compare heights within a community type to see strengths/weaknesses
- Compare across community types to see gaps

Why it matters: Identifies which dimension needs most improvement in each community type.



Population Distribution

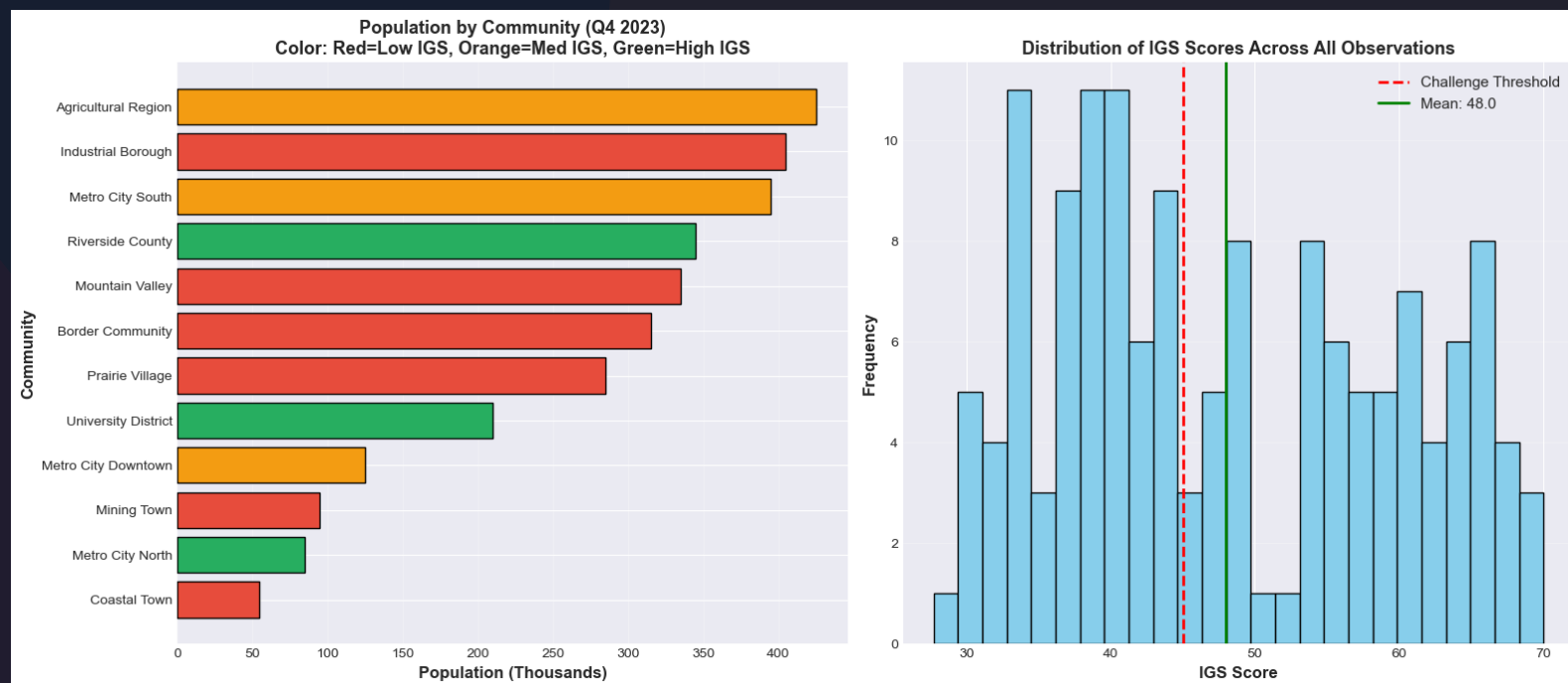
What it shows: Two panels showing:

1. **Left:** Population of each community (colored by IGS category)
2. **Right:** Distribution of all IGS scores in the dataset

How to read it:

- Bar colors: Red (Low IGS), Orange (Medium), Green (High)
- Histogram shows how many observations fall in each IGS range

Why it matters: Shows if low-IGS communities serve large populations (more urgent intervention needed).



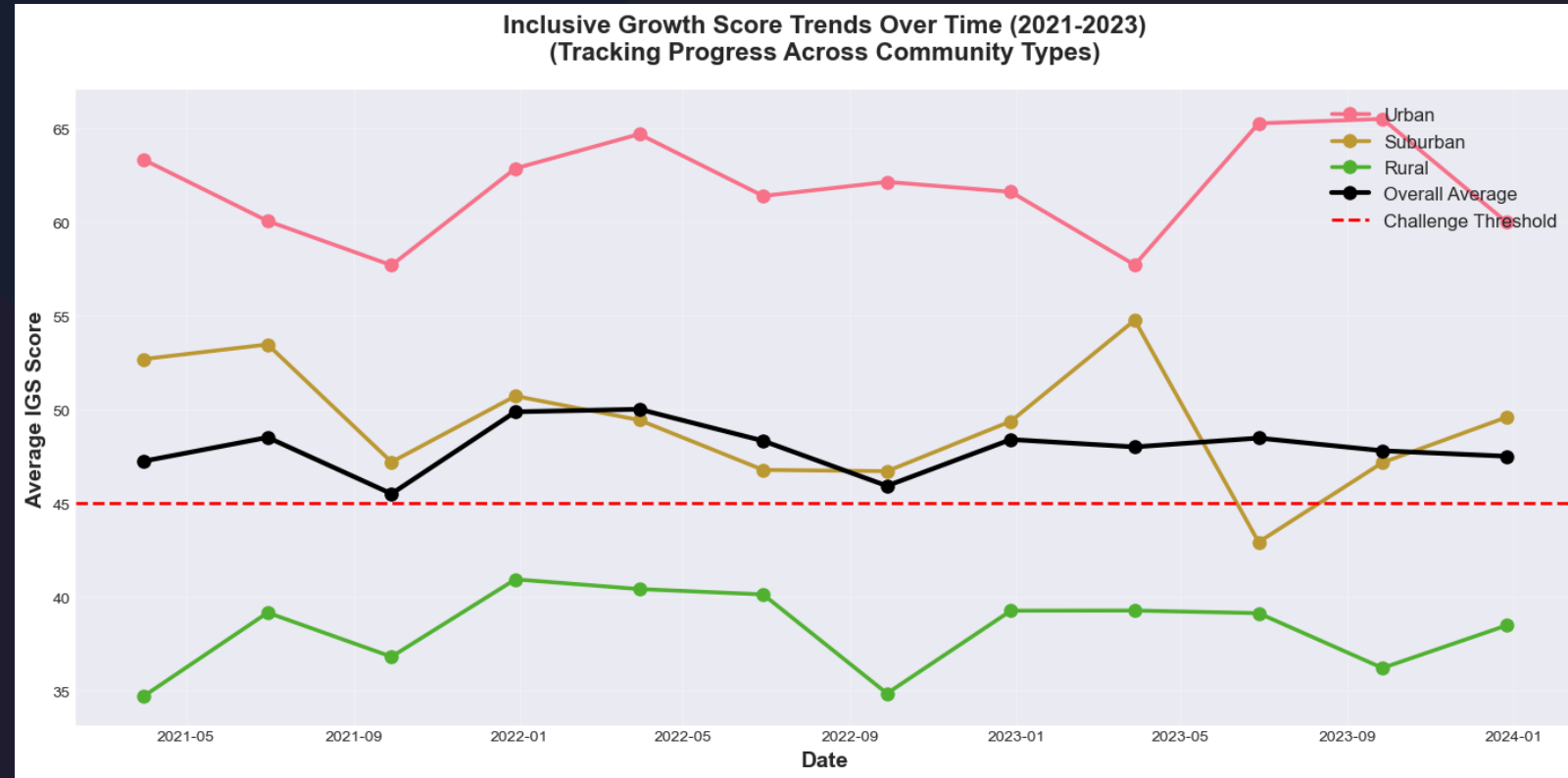
Time Series

What it shows: How IGS scores change over time for different community types.

How to read it:

- X-axis = time (quarters from 2021 to 2023)
- Y-axis = average IGS score
- Each line = one community type
- Red dashed line = threshold (45)

Why it matters: Shows if communities are improving, stagnating, or declining.



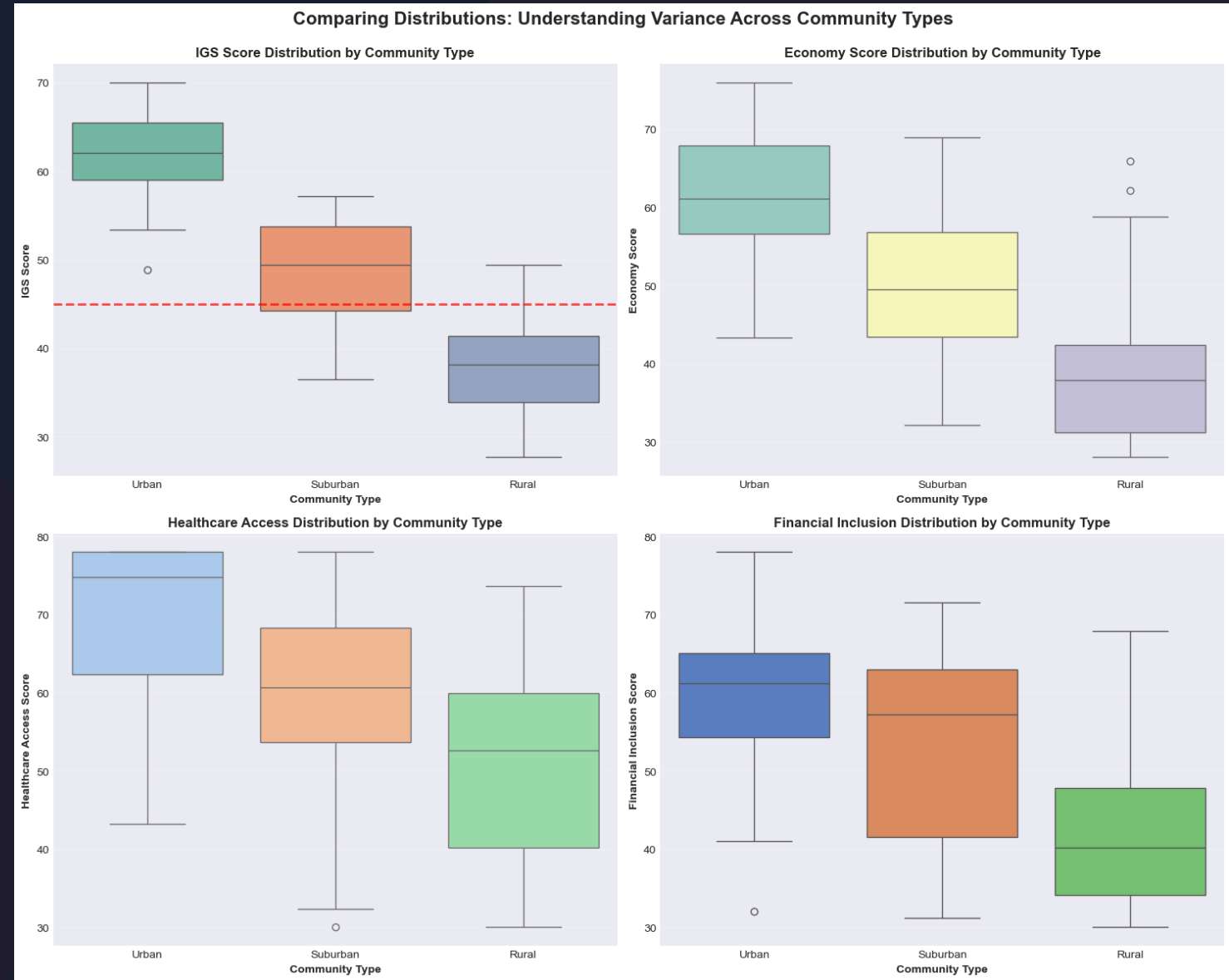
Box Plots - Distribution Comparison

What it shows: Distribution of IGS, Economy, Healthcare, and Financial Inclusion across community types.

How to read a box plot:

- Box = middle 50% of data (25th to 75th percentile)
- Line inside box = median (50th percentile)
- Whiskers = range of data (excluding outliers)
- Dots = outliers

Why it matters: Shows not just averages, but the full spread and variability within each community type.



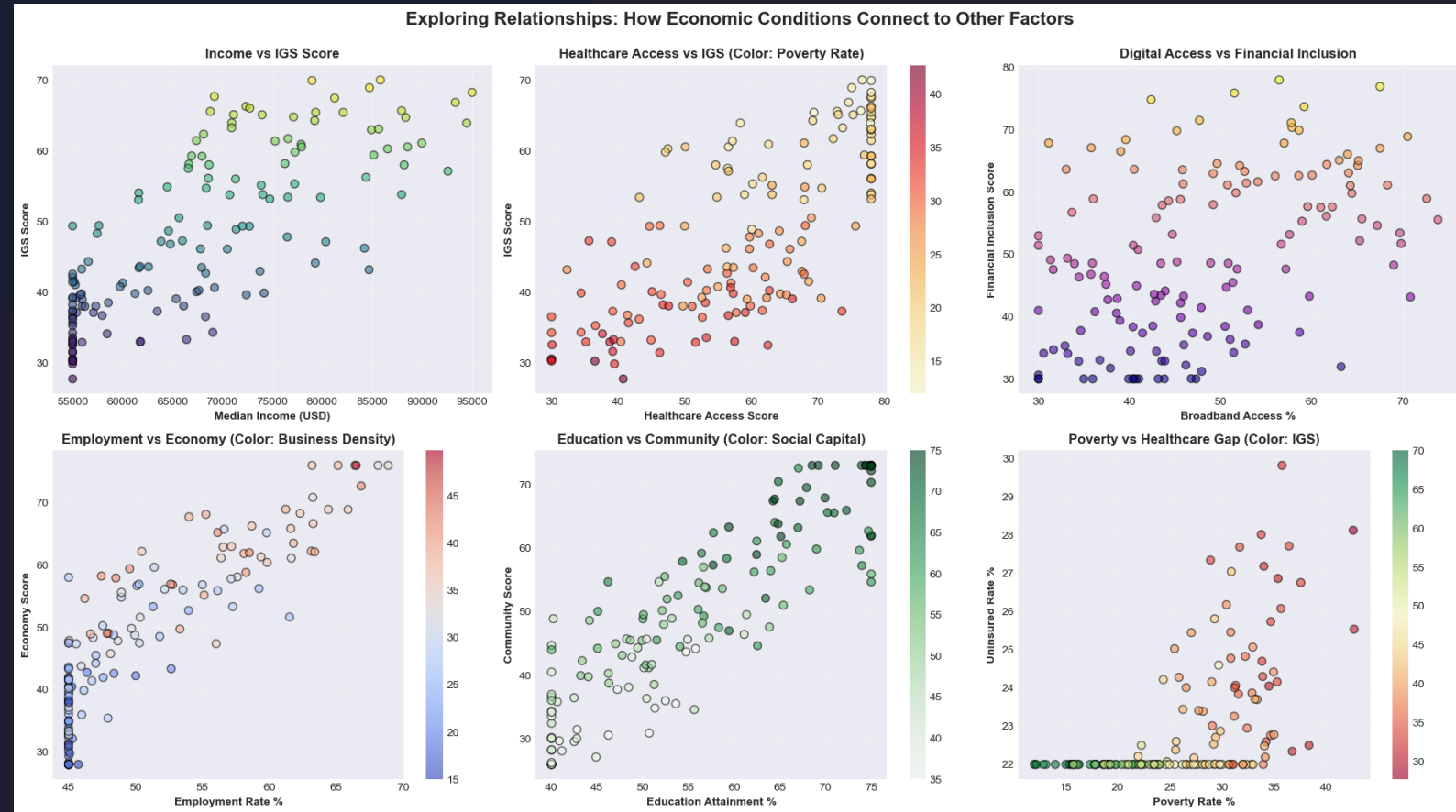
Scatter Plots

What it shows: Six scatter plots showing relationships between key variables.

How to read scatter plots:

- Each dot = one observation (community in one quarter)
- Upward slope = positive relationship
- Downward slope = negative relationship
- Tight clustering = strong relationship
- Scattered dots = weak relationship

Why it matters: Visual proof of which factors drive IGS and which communities have specific challenges.



Statistical and Modeling Techniques



Basic Statistical Analysis

- Descriptive Stats: Mean, Median, Mode
Mode
- Significance Tests: t-score, p-value, z-score
- Correlations & Relationships
Relationships



Basic Statistical Models

- Linear Regression (Baseline)
(Baseline)
- Logistic Regression (Classification)
- Hypothesis Testing Frameworks
Frameworks



Foundations of AI Models

- Decision Trees & Random Forests
- Ensemble Methods (Gradient Boosting)
- Intro to Neural Networks



Explainable AI (XAI)

- Feature Importance Analysis
- Model Diagnostics (SHAP, LIME)
- Understanding Limitations & Bias

OUR GOAL

1 What happened?



2 What's driving it?
it?



3 What should we do?

Statistical Testing

Statistical tests help us answer questions like:

- "Is the difference between Urban and Rural IGS **real** or just **random chance**?"
- "Is there a **statistically significant** relationship between income and IGS?"

Key Concepts:

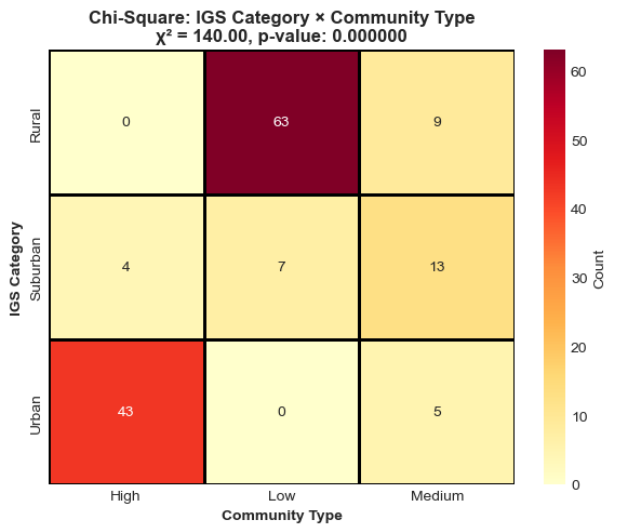
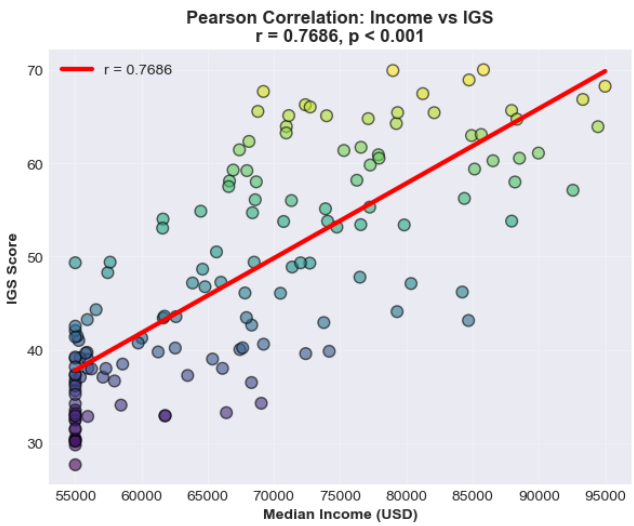
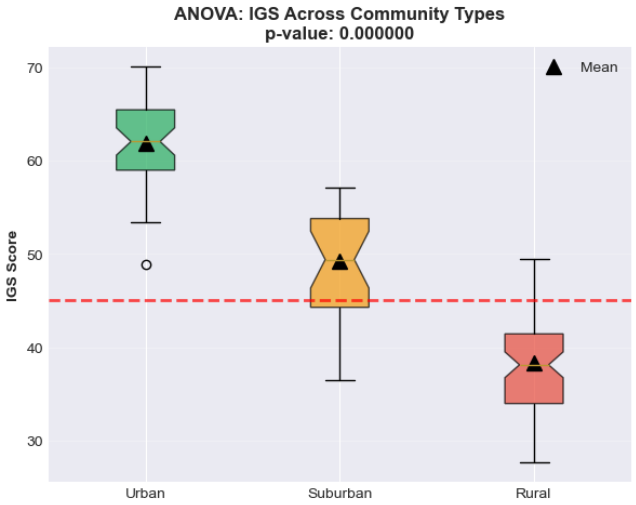
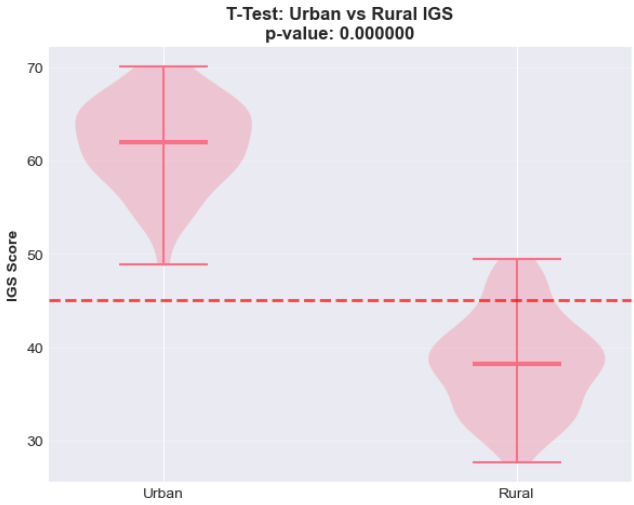
1. **Null Hypothesis (H_0)**: The "boring" hypothesis that there's no difference or no relationship
2. **Alternative Hypothesis (H_1)**: The hypothesis that there IS a difference or relationship
3. **p-value**: Probability that we'd see these results if H_0 were true
 - $p < 0.05 \rightarrow$ We **reject** H_0 (result is statistically significant!)
 - $p \geq 0.05 \rightarrow$ We **cannot reject** H_0 (might be random chance)

Tests We'll Run:

1. **T-test**: Compare two groups (Urban vs Rural IGS)
2. **ANOVA**: Compare three or more groups (Urban vs Suburban vs Rural)
3. **Pearson Correlation**: Test if two continuous variables are related
4. **Chi-Square**: Test if categorical variables are associated

Statistical Test Visualizations

Statistical Tests: Understanding Group Differences and Relationships



Machine Learning Models

What is Machine Learning?

Machine learning is teaching computers to **learn patterns from data** rather than explicitly programming rules.

Our Goal:

Build models that can **predict IGS scores** based on other variables (Place, Economy, Community scores, income, healthcare access, etc.)

Why This Matters:

1. Understanding drivers: Which variables have the most impact on IGS?
2. Prediction: Can we predict IGS for new communities?
3. Policy testing: What happens if we improve healthcare by 10 points?

Models We'll Build:

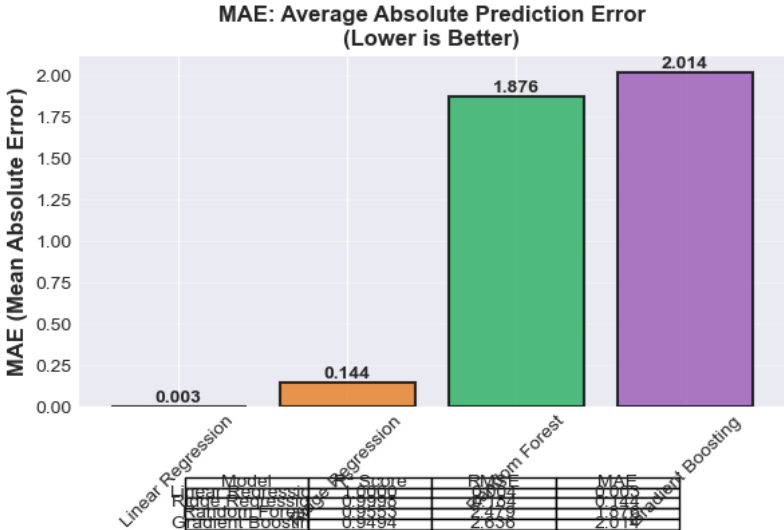
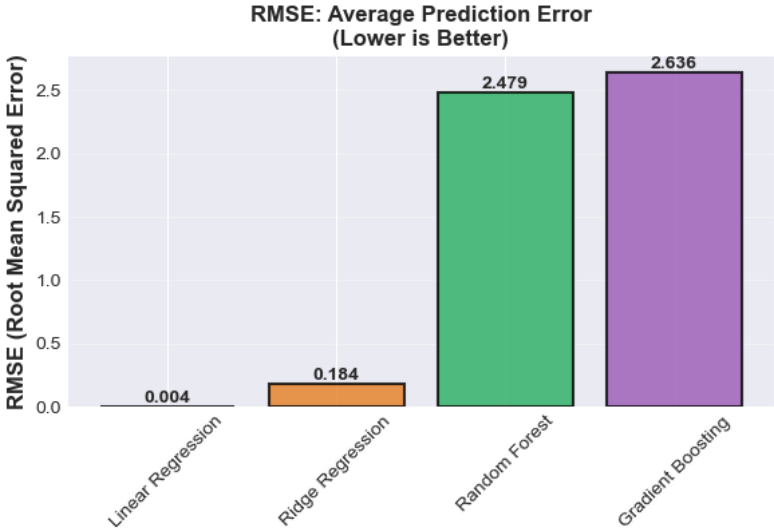
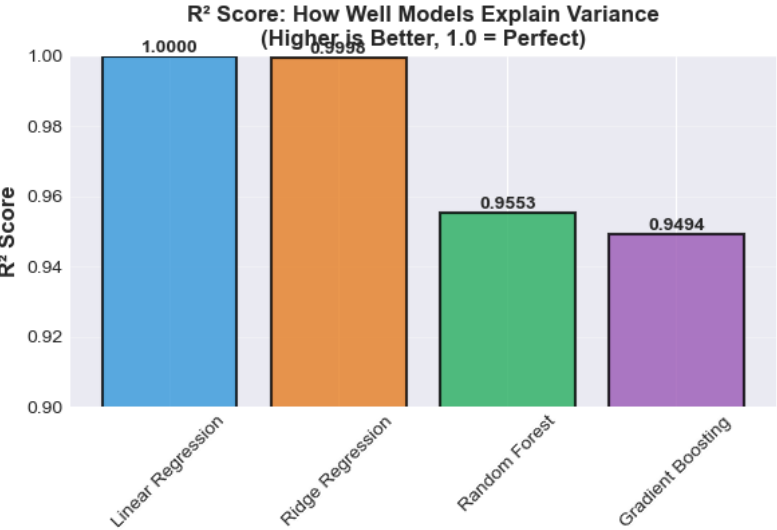
1. **Linear Regression**: Simple, interpretable baseline
2. **Ridge Regression**: Linear model with regularization (prevents overfitting)
3. **Random Forest**: Ensemble of decision trees (handles complex patterns)
4. **Gradient Boosting**: Sequential tree building (often highest accuracy)

Evaluation Metrics:

- R^2 Score: How much variance is explained? (0 to 1, higher is better)
- RMSE: Root Mean Squared Error (average prediction error in IGS points)
- MAE: Mean Absolute Error (average absolute prediction error)

Model Performance Comparison

Machine Learning Model Performance Comparison



Performance Summary Table

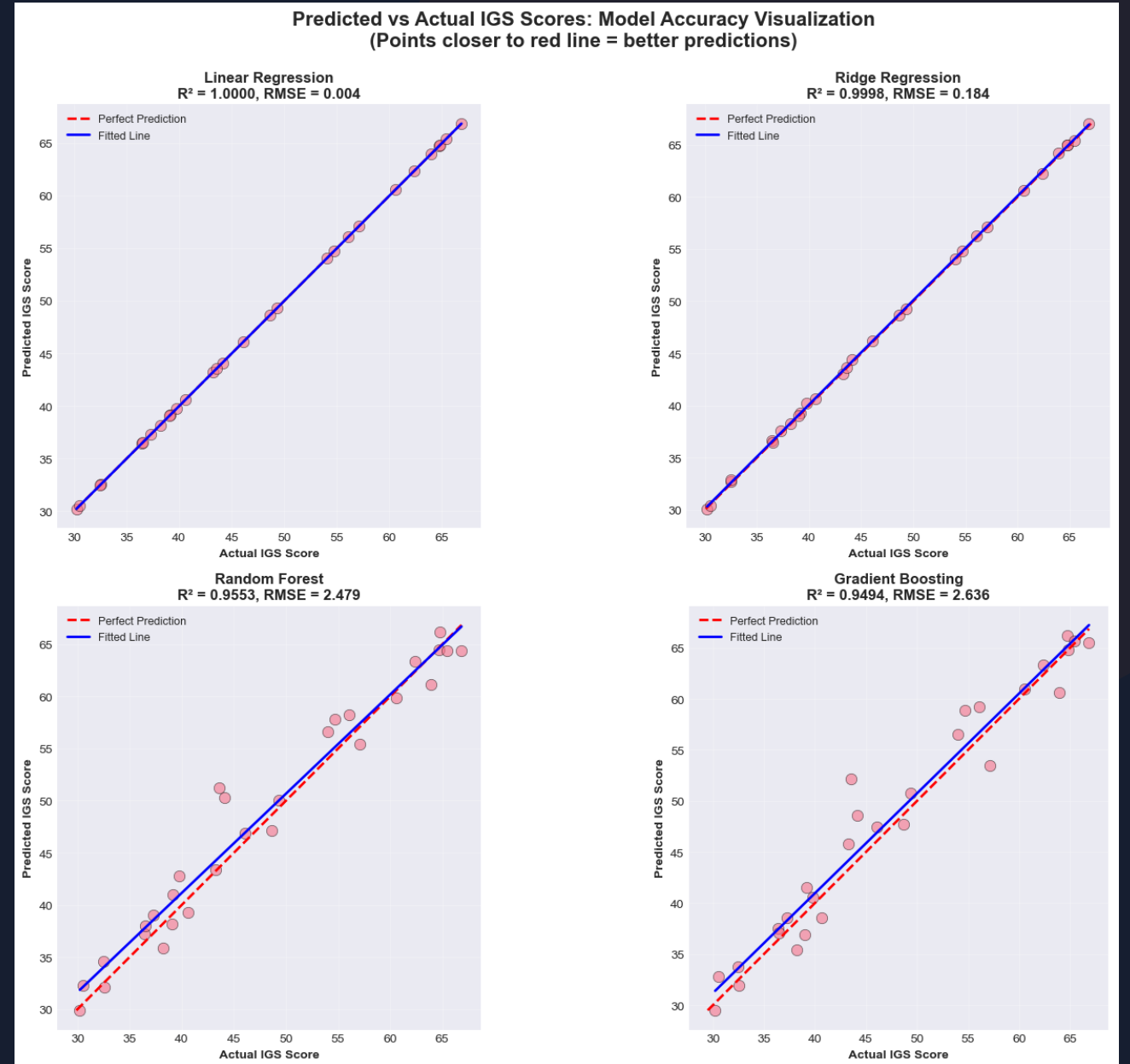
Predicted vs Actual Values

What it shows: How close our model predictions are to actual IGS scores.

How to read it:

- Each dot = one test observation
- X-axis = actual IGS score
- Y-axis = model's predicted IGS score
- Red dashed line = perfect predictions
- Blue solid line = actual fit

Why it matters: Shows if model is accurate and if there are systematic errors.



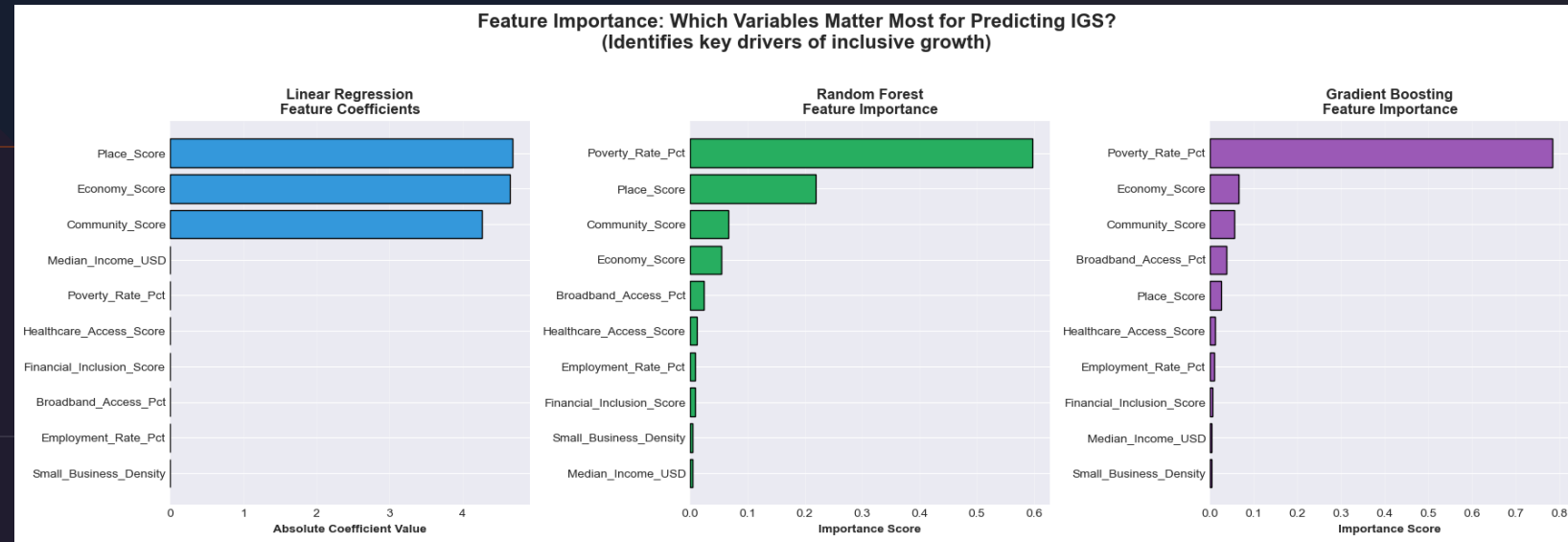
Feature Importance

What it shows: Which variables are most important for predicting IGS.

How to read it:

- Longer bars = more important features
- Shows which factors have the biggest impact on IGS

Why it matters: Tells policymakers which areas to focus on for maximum IGS improvement.



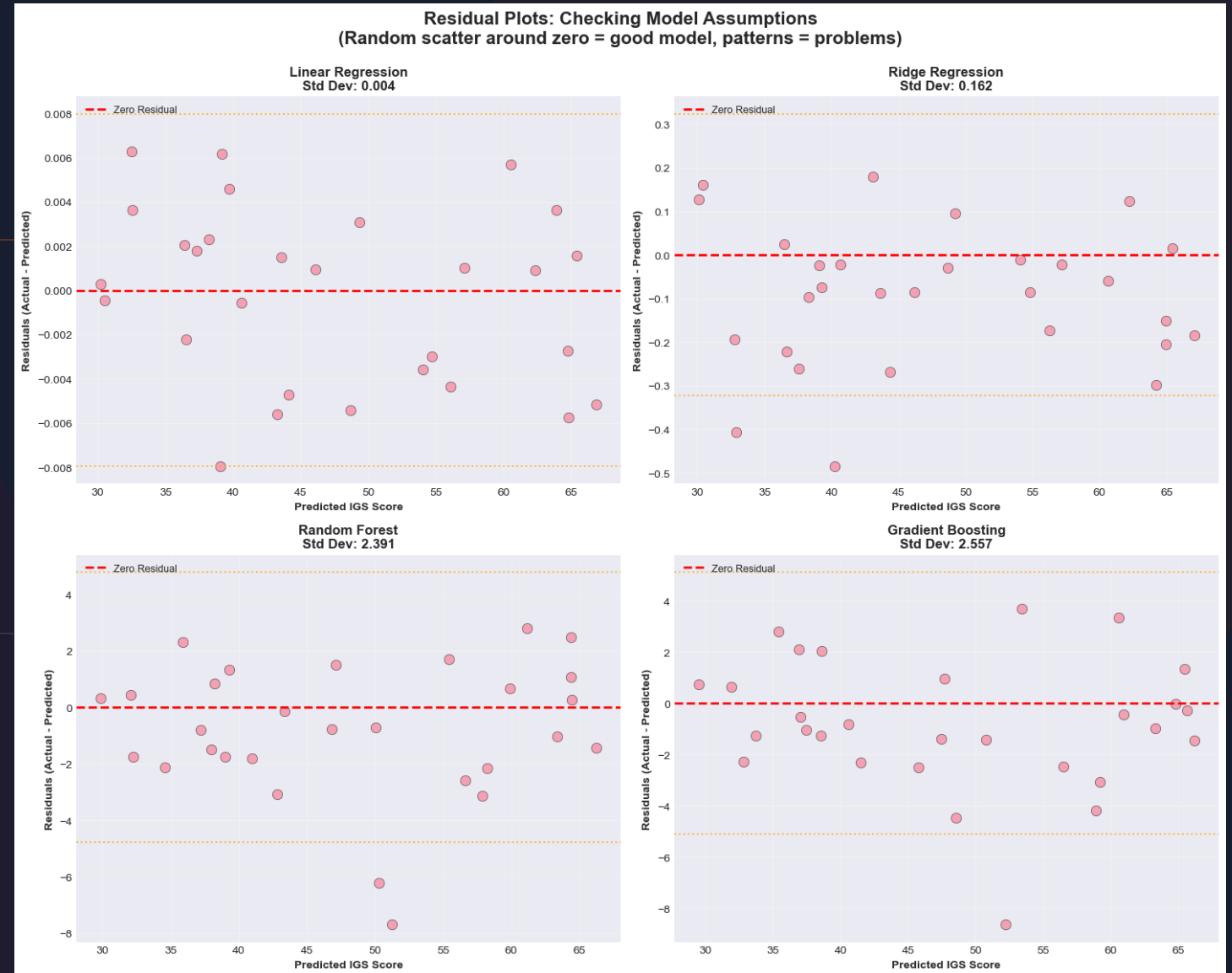
Residual Analysis

What it shows: Pattern of prediction errors.

How to read it:

- X-axis = predicted IGS score
 - Y-axis = residual (actual - predicted)
 - Residual = 0 means perfect prediction
- Random scatter around zero = good model
- Pattern or trend = model problem

Why it matters: Validates model assumptions and identifies systematic prediction errors.



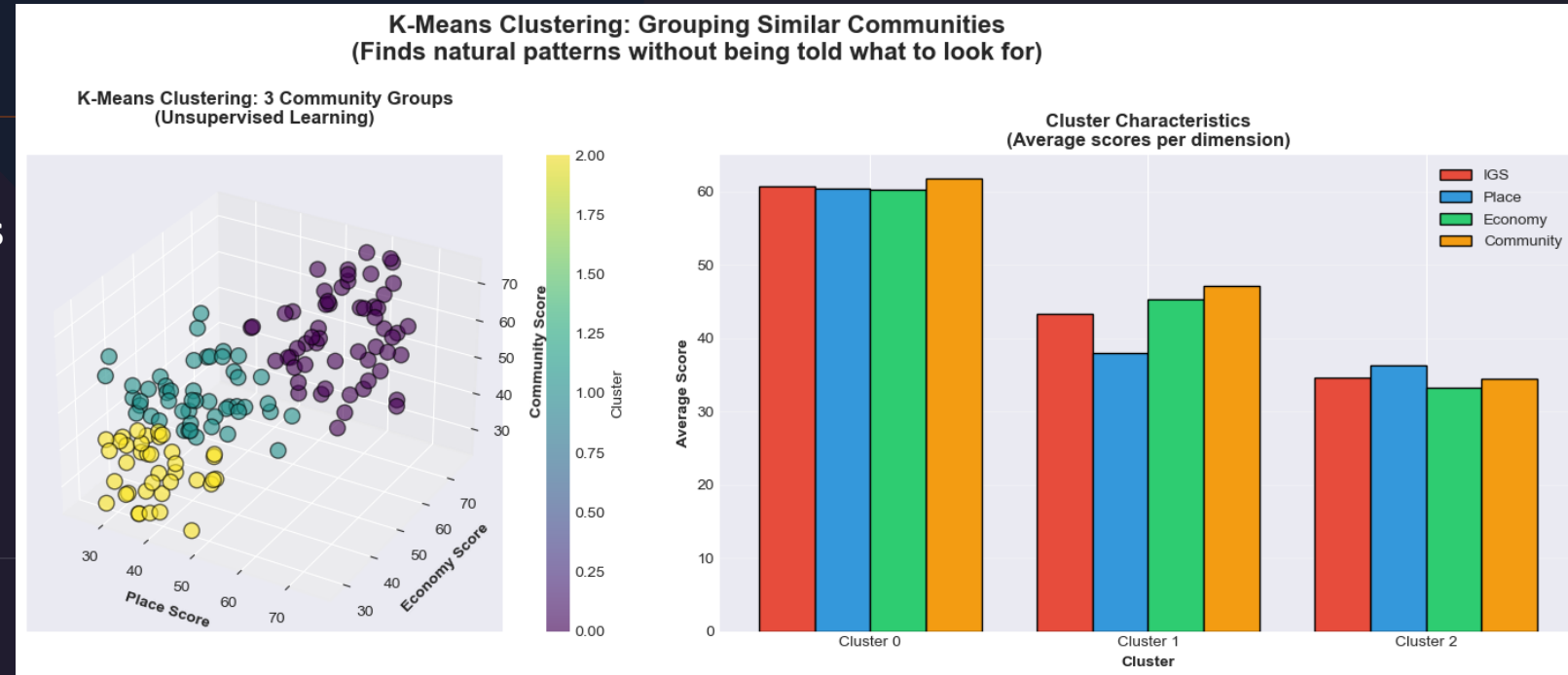
K-Means Clustering

What it shows: Communities grouped by similarity across all indicators.

How to read it:

- Communities with similar characteristics are in the same cluster
- 3D plot shows Place, Economy, and Community scores
- Bar chart shows average scores for each cluster

Why it matters: Helps identify distinct community profiles for targeted policies.



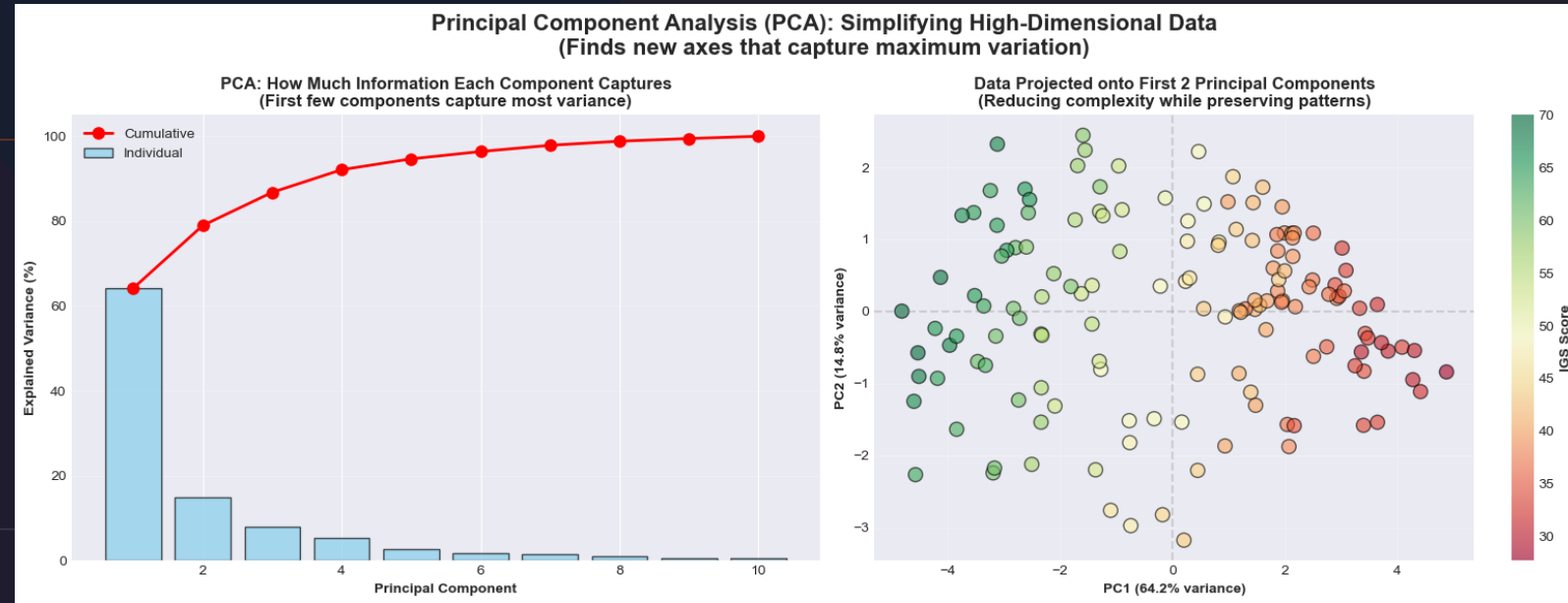
Principal Component Analysis (PCA)

What it shows: All our data compressed into 2 main dimensions.

How to read it:

- Left panel: How much information each component captures
- Right panel: All observations plotted on first 2 principal components
- Color = actual IGS score

Why it matters: Shows if our 10+ features can be summarized by a few key patterns.



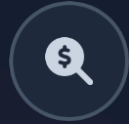
Turning Analysis into Insight

The Insight Workflow



Target Population

Filter for tracts with IGS < 45 (Low Growth)



Hypothesize

Use EDA to identify potential drivers



Confirm

Use statistical models to quantify impact



Recommend

Design measurable interventions

Example Intervention Logic Models



Place Intervention

INPUT (DRIVER)

Broadband + Transit Access



INTERMEDIATE OUTCOME

↑ Financial Inclusion



FINAL IMPACT

↑ Small Business Activity



Economy Intervention

INPUT (DRIVER)

Workforce Dev + Capital



INTERMEDIATE OUTCOME

↑ Employment & Income



FINAL IMPACT

↑ Economic Resilience



Community Intervention

INPUT (DRIVER)

Health & Education Access



INTERMEDIATE OUTCOME

↑ Civic Participation



FINAL IMPACT

↑ Community Stability

Thank You and Q&A



Open Discussion

Now is the time to ask about specific data data challenges, tool limitations, or clarification clarification on the IGS methodology.

Helpful Resources

-  Mastercard Center for Inclusive Growth Growth
-  Python Notebooks (GitHub)
-  dsi-support@auc.edu
support@auc.edu